

Spherical Harmonics

EE5178 | 3D Gaussian Splatting Super-Resolution

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Baseline Pipeline

COLMAP (various arguments) → Gaussian Splatting (Kerbl et al.) → Render Test Views → 4× Upscale (Linear Interpolation)

1. COLMAP (Command Line)



Input: Image Collection

Example command line methods
(various arguments tried)

```
# Exhaustive matcher (baseline)
colmap feature_extractor \
  --database_path database.db --ImageReader.single_camera 1
colmap exhaustive_matcher \
  --database_path database.db
```

```
# Sequential matcher
colmap sequential_matcher \
  --database_path database.db \
  --SequentialMatching.overlap 10
```

```
# Vocab tree matcher
colmap vocab_tree_matcher \
  --database_path database.db \
  --VocabTreeMatcher.num_images 6
```

```
# Mapper (common for all)
colmap mapper \
  --database_path database.db \
  --image_path images \
  --output_path sparse
```



Output:
Sparse Reconstruction
(cameras, points)

2. Gaussian Splatting

(Kerbl et al. – 3D Gaussian Splatting)



Input:
COLMAP Sparse
Reconstruction
(cameras, points)



Train Gaussian Splatting
Model
(kerbl/gaussian-splatting)



<https://github.com/graphdeco-inria/gaussian-splatting>

3. Render Test Views & 4× Upscale

Render Novel Test Views (1x)



View 1 (1x)



View 2 (1x)

...

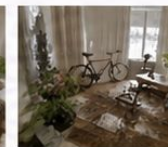


View N (1x)

4× Upscale by Linear Interpolation



View 1 (4x)



View 2 (4x)

...



View N (4x)

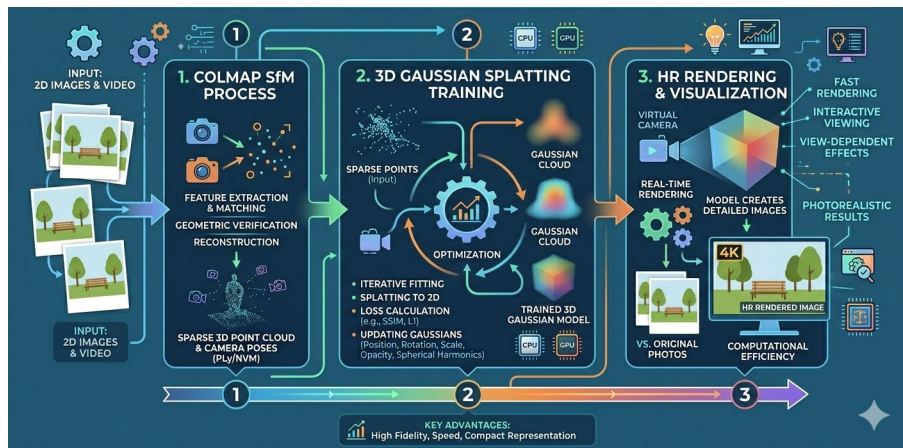
Upscaling Method:
Linear Interpolation (factor = 4x)



Baseline Summary

COLMAP (multiple command-line configurations) → 3D Gaussian Splatting (Kerbl et al.) → Render novel test views → 4× upscale using linear interpolation

Method-1: 3D Gaussian Splatting (Path A)



- Pipeline: **COLMAP SfM** → **3DGS Training** → **HR Rendering**
- Point cloud initialization from COLMAP Structure-from-Motion
- **15,000** training iterations with L1 + **SSIM*** combined loss
- AbsGS densification: 2D screen-space gradient for better Gaussian placement [Ye et al., 2024]
- SA-GS scale-adaptive anti-aliasing filter ($\text{eps}_{2d} = 0.3 \times r^2$) [Song et al., CVPR 2024]
- Per-view affine color correction: NeRF-W* style, 6 params per camera [Martin-Brualla et al.]
- Render novel views at 4x HR resolution directly from trained Gaussians

◆ Applied to scenes with good colmap reconstruction: Bike, Bbuddha, Firehydrant

◆ Key innovation: Scene-adaptive routing

COLMAP reconstruction quality determines whether to use 3DGS or SR fallback. This single decision contributed **+0.19** to final score.

*SSIM- The Structural Similarity Index (SSIM)

*NeRF-W - Neural Radiance Fields in the Wild



What Didn't Work

Transparent reporting of negative results

Approach	Expected	Actual	Root Cause
LPIPS loss (0.1)	Better quality	0.734 (−0.03)	Fights SSIM/PSNR metric
SRGS cross-scale	+0.02–0.04	0.649–0.692	Hurt loss convergence
Per-pixel fusion	Best of both	0.554	Alpha = coverage ≠ quality
MiDaS depth reg	Better geometry	Worse	Noisy pseudo-depth, sparse views
Test-Time Opt	+0.01	+0.0007	Negligible on sparse data
Unsharp mask	Sharper edges	−0.003	Artifacts degrade PSNR

Method-2: HAT Super-Resolution (Path B)

HAT MODEL & FEATURES



HAT: Hybrid Attention Transformer

[Chen et al., CVPR 2023]



Direct 2D Image Upscaling
(no 3D reconstruction needed)



8-way Geometric Self-Ensemble: Flip + rotate augmentations, average predictions



Replaced Real-ESRGAN Baseline with HAT for +0.006 improvement



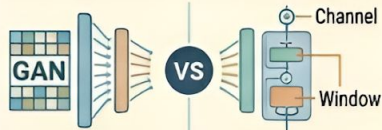
Robust fallback for scenes where COLMAP reconstruction fails

Designed for high-fidelity, artifacts-free 4× SR

SR MODEL COMPARISON: WHY HAT?

HAT over Real-ESRGAN

capturing both local texture and global structure better than Real-ESRGAN's GAN-based approach



Real-ESRGAN:
GAN-based

HAT: Transformer
w/ Hybrid Attention



Local Texture



Global Structure



Superior detail preservation

APPLIED SCENES (HAT SR)



Applied to Scenes with Poor Reconstruction



COLMAP cameras

Scene	Method	Issue
Aeroplane	HAT	Low texture, poor COLMAP poses.
Cycle	HAT	Occlusion and complex structure.
Face	HAT	Artifacts, pose estimation failure.
Still3	HAT	Thin structures, poor sparse point cloud.
Toy	HAT	Fine detail, pose initialization failure.

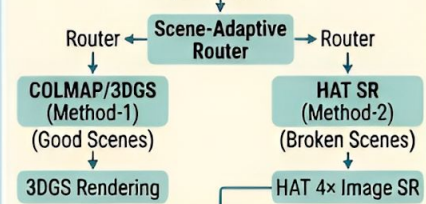
Targeted scene processing for optimal quality

OUR KEY CONTRIBUTION: SCENE-ADAPTIVE ROUTING



Our Contribution vs. One-Size-Fits-All

Original Scene Data (2D Images)



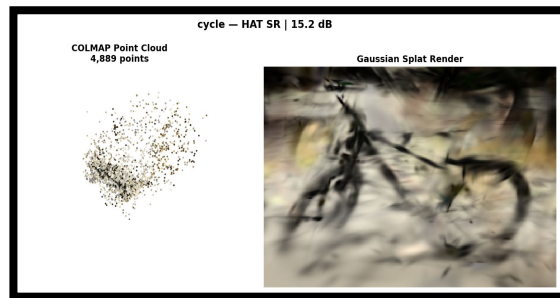
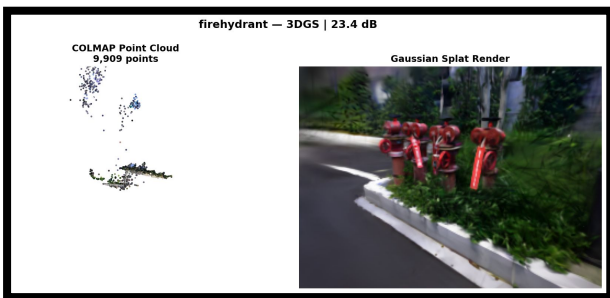
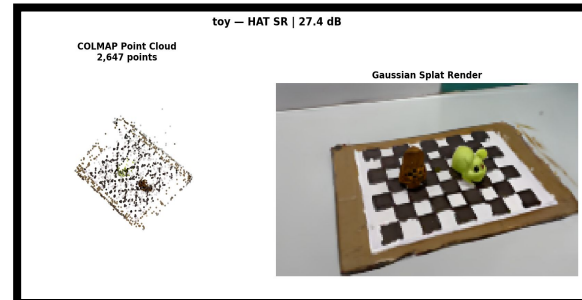
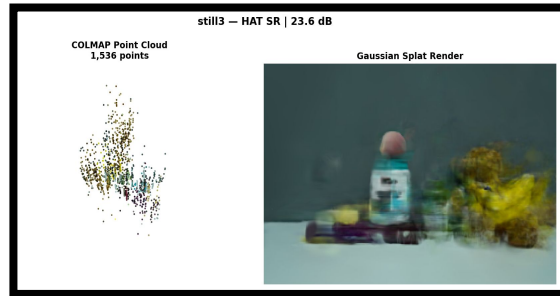
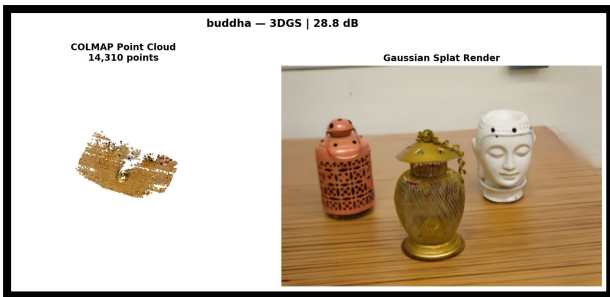
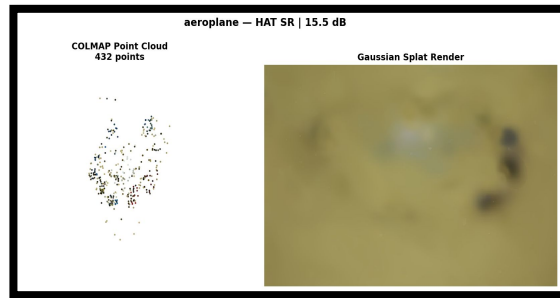
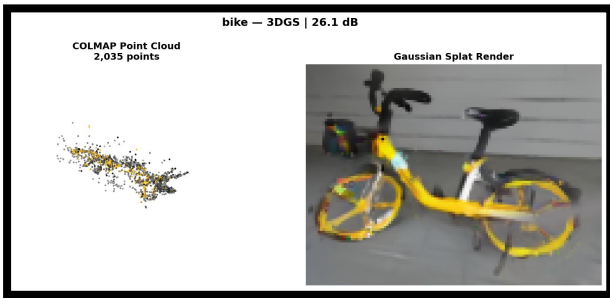
Optimized Pipeline Output (Final HR Image)



Our routing selects 3DGS (Method-1) for good scenes and HAT SR (Method-2) for broken scenes, instead of a one-size-fits-all model like Real-ESRGAN.

Maximized overall pipeline fidelity

Input Point Cloud and Output Gaussian Splat — All Scenes



LR INPUT

aeroplane — LR Input



Aero
Plane

SR OUTPUT

aeroplane — SR Output (HAT SR)



bike — LR Input



Bike

bike — SR Output (3DGS)



buddha — LR Input



Buddha

buddha — SR Output (3DGS)



**Qualitative Results: LR
Input vs SR Output (4x)
- All 8 Scenes**

LR — Low Resolution
SR — Super Resolution

LR INPUT

cycle — LR Input



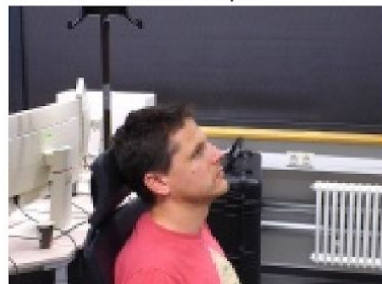
Cycle

SR OUTPUT

cycle — SR Output (HAT SR)



face — LR Input



Face

face — SR Output (HAT SR)



firehydrant — LR Input



FireHydrant

firehydrant — SR Output (3DGS)



**Qualitative Results: LR
Input vs SR Output (4x)
- All 8 Scenes**

LR — Low Resolution
SR — Super Resolution

Quantitative results of Novel views

